MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 3SP1

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus, rectangular

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Alt
Unit:	V	Vdc	v	Vdc	Vdc	Meg	Meg	ft
Maximum:	6.9	0, -200	550	1,100	2,750	1.5	1	10,000
Minimum:	5.7	---	---	---	500	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	1,500	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(E) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	D1 to all D2 to all D3 to all D4 to all	CD1 CD2 CD3 CD4	--- --- --- ---	19.0 15.0 14.0 15.0	pF pF pF pF
(E) 5248	Deflection-factor uniformity	-		---	---	---	---
5216	Cathode illumination	-		---	---	---	---
5246	Focusing voltage at modulation condition	-		Eb1	247	---	Vdc
(E) 5111	Vibration	-		---	---	---	---
	Quality conformance inspection, part 1						
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1, 2	Light = 5 ftL	---	---	---	---
5221	Light output	1, 2	Eb2 = 1,500 Vdc	Light	5.0	---	ftL
5223	Modulation	2	Light = 5 ftL	ΔEc	---	48.0	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	12.0	mm
5231	Spot displacement (leakage)	-		Displ	---	7.0	mm
5241	Grid cutoff voltage	-		Eco	-43.5	-101	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode currents (anode)	2	Ec1 = 0; light = 5 ftL	Ib1	-15	+10	μAdc
(E) 1101	Secureness of base, cap, or insert	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 - Continued						
5216	Stray light emission (conventional types)	-	Eb2 = 2,750 Vdc	---	---	---	---
5226	Line width A	2	Eb2 = 1,500 Vdc; light = 5 ftL	Width	---	0.4	mm
5226	Line width B	2	Eb2 = 1,500 Vdc; light = 5 ftL	Width	---	0.4	mm
(E) 5101	Trace and bulb alinement (rectangular electrostatic types)	-	1D2, bulbwall	---	---	---	---
5246	Focusing voltage at cutoff	2	Light = 5 ftL	Eb1	---	465	Vdc
5248	Deflection factor	-	1D2	DF	109	149	Vdc/in.
5248	Deflection factor	-	3D4	DF	78.0	105	Vdc/in.
5251	Heater-cathode leakage current	-		---	---	---	---
(E) 5101	Base alinement (electrostatic types)	-	+1D2; keyway	---	---	---	---
(E) 5101	Neck and bulb alinement (electrostatic types)	-		---	---	---	---
(E) 5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
(E) 5101	Angle between traces	-		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	2	Group C; light = 5 ftL; Eb2 = 2,750 Vdc	t	500	---	hrs
---	Life-test end points:		Light = 4.25 ftL				
5226	Line width A	-		Width	---	0.4	mm
5226	Line width B	-		Width	---	0.4	mm
5223	Modulation	-		ΔEc	---	48.0	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
5251	Heater-cathode leakage current	-		---	---	---	---
5216	Stray light emission (conventional types)	-		---	---	---	---

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NOTES:

1. This test to be performed at the conclusion of the holding period.

2. Use a raster size of 1.125 by 2.750 inches (28.58 mm by 69.85 mm) and a type three photronic cell with eye correction, stopped down to a 1-inch (25.40 mm) aperture, and calibrated in foot candles of illumination from a light source having a color temperature of 2,700°K.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

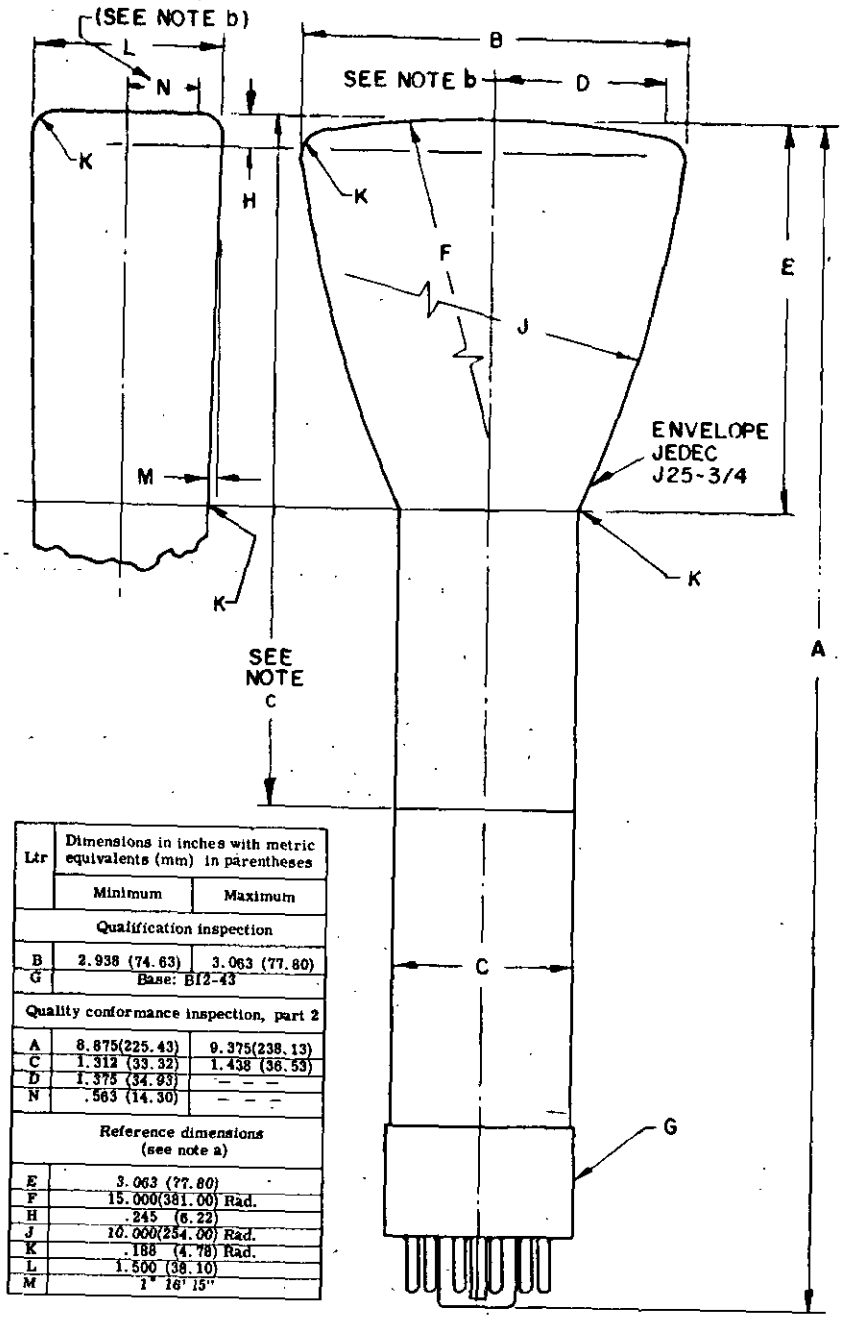
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

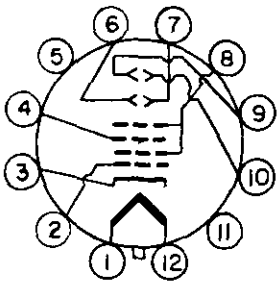
Agent: DSA - ES

(Project 5960-2645)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	2.938 (74.63)	3.063 (77.80)
G	Base: B12-43	
Quality conformance inspection, part 2		
A	8.875(225.43)	9.375(238.13)
C	1.312 (33.32)	1.438 (36.53)
D	1.375 (34.93)	- - -
N	.563 (14.30)	- - -
Reference dimensions (see note a)		
E	3.063 (77.80)	
F	15.000(381.00) Rad.	
H	.245 (6.22)	
J	10.000(254.00) Rad.	
K	.188 (4.78) Rad.	
L	1.500 (38.10)	
M	1° 16' 15"	



BASING DIAGRAM

NOTES:

a. These dimensions are for information only and are not required for inspection purposes.
b. The minimum useful screen area shall be a rectangle 2.75(69.85 mm) by 1.125(28.58 mm) centered at the geometric center of the face.
c. This portion of tube envelope shall be coated with clear baking synthetic XS-1810, as supplied by Coating Materials Laboratories, or equivalent.

FIGURE 1. Outline drawing of electron tube type 3SP1.

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